Thomas C. Fraser

Curriculum Vitae
November 15, 2024

Personal

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Address	Strynøgade 5, 1.08, 2100 Copenhagen, Denmark
Citizenship	Canadian
Date of Birth	27/11/1994

Positions & education

Jan. 2024 – present	Postdoctoral Researcher *Department of Mathematics, University of Copenhagen*
Sept. 2018 – Sept. 2023	Ph.D., Physics Thesis: An estimation theoretic approach to quantum realizability problems *Perimeter Institute For Theoretical Physics, Waterloo, ON*
Aug. 2017 – Jun. 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility *Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON*
Sept. 2012 – Apr. 2017	B.Sc., Mathematical Physics, Astrophysics Specialization *University of Waterloo, Waterloo, ON*

Papers

2023 Thomas. C. Fraser
"An estimation theoretic approach to quantum realizability problems"
Ph.D. Thesis. arXiv:2402.10902

2023 D. Schmid, Thomas. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
"Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory"
Quantum 7, 1194

2022 Thomas. C. Fraser
"A sufficient family of necessary inequalities for the compatibility of quantum marginals"
arXiv preprint arXiv:2211.00685

2020 Thomas. C. Fraser
"A Combinatorial Solution to Causal Compatibility"
Journal of Causal Inference 8(1), 22-53

2018 J. R. Rinehart, J. H. Béjanin, Thomas. C. Fraser, M. Mariantoni
"Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry"
arXiv preprint arXiv:1804.04756

2018 Thomas. C. Fraser, E. Wolfe
"Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario"
Physical Review A, 98(2), 022113

Teaching

Block 3 2023	Introduction to Representation Theory - University of Copenhagen *Teaching Assistant*
Winter 2021	Quantum Information - Perimeter Scholars International (PSI) Program *Teaching Assistant*
Winter 2021	Quantum-classical gaps in causal structures - PSI Winter School *Student project co-lead*
Fall 2018	Physics 1 - Phys 111 - University of Waterloo *Teaching Assistant*

Talks

2022	The IQC-QuICS Math and Computer Science Seminar *A sufficient family of necessary inequalities for the quantum marginals problem*
2021	Beyond IID in Information Theory 9 *Simultaneous Spectral Estimation and the Quantum Marginal Problem*
2021	PI Grad Student Conference, Waterloo, Canada *Simultaneous Spectral Estimation and the Quantum Marginal Problem*
2019	IQST Seminar at University of Calgary, Canada *Sudoku on steroids - The combinatorics of causal inference*
2016	Quantum Networks Conference at International Institute for Physics, Natal, Brazil *Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario*

Non-academic Experience

September 2015 – January 2016	*Research & Development Data Scientist* Sysomos, Toronto, ON, Canada
September 2014 – May 2015	*Video Game Developer* Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	*Mathematics Tutor* Humber College, Toronto, ON, Canada
June 2011 – August 2011	*Solar Panel Technician* OVG Solar, Inc, Renfew, ON, Canada

Grants & Awards

2022 *Natural Sciences and Engineering Research Councilof Canada (NSERC)*
Three years, Grant No. 411301803

2018 *UW Graduate Award & Marie Curie Award*
Sufficiently High Entrance Average, Department Selected

2018 *The Joanne Cuthbertson and Charlie Fischer Graduate Student Award*
Support for an exceptional Masters student from the 2017/2018 PSI class

2017 *Governor General's Silver Academic Medallion*
Highest Graduating Average in Graduating Class

2016 *Mike Lazaridis Scholarship*
Theoretical Physics Fellowship at Perimeter Institute

2015 *Xerox Research Centre of Canada Limited Award*
Best work-term report "Acoustic Modelling Using Mel-Frequency Cepstral Coefficients"

2015 *C. C. Lim Physics Prize*
Top Marks in Undergraduate Thermodynamics

2013 *Don E. Brodie Scholarship*
Highest Experimental Physics Lab Performance

2012 *A. Donald Maynes Scholarship*
Outstanding Academic Record

2012 *BMO Undergraduate Entrance Scholarship*
Outstanding Academic Average